

COURSE: “CYBER FORENSICS AND ETHICAL HACKING”

TOPIC 1

WINDOWS OS SECURITY

Lesson 1 · Windows OS Services | Lecture, 2 hours



REPUBLIC OF UZBEKISTAN
UNIVERSITY OF MILITARY SECURITY AND
DEFENSE
*Institute of ICT and Military Communications ·
Cybersecurity Department*

Lecturer: Senior Lecturer, Cybersecurity Department — Captain M. Isakov

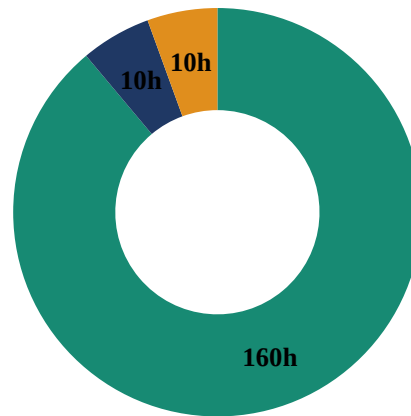
Classroom hours — distribution

180

classroom hours

Taught in semesters 5, 6 and 7
(60 hours each)

By type (180 hours total)



■ Practical ■ Lecture ■ Seminar



What is a classroom session?

Sessions taught in class with an instructor: lecture, practical and seminar.

Main focus — **practical sessions (160 hours):** the cadet applies theory in a real/virtual environment.

Each semester — 60 classroom hours; 180 hours across 3 semesters.

Note: there is also independent study (150 hours), but lessons are delivered specifically in the classroom hours.

Course goal and expected outcomes



GOAL — what we want to achieve

To train the cadet as a cybersecurity officer who understands Windows and Linux systems from the inside, can find malicious software, track digital traces in networks and systems, and can carry out both attack and defense in practice.

In short: teaching not “which button to press”, but how the system works and how to protect it.



EXPECTED OUTCOME — what the cadet will be able to do

- ▶ Manage a Windows/Linux system at administrator level
- ▶ Recognize and analyze viruses and malicious code
- ▶ Know and apply SIEM, EDR, DLP, IPS/IDS defense systems
- ▶ Work with Metasploit and network scanners
- ▶ Collect information from open sources via OSINT

Course structure — 3 semesters, 6 topics

Semester 5	60 hours	Semester 6	60 hours	Semester 7	60 hours
	Topic 1. Windows OS security		Topic 4. Network monitoring & security systems (DLP, IDS/IPS, SIEM, EDR, Metasploit)		Topic 5. Cyber intelligence & ethical hacking (OSINT, C2, forensics)
	Topic 2. Linux OS				Topic 6. Project development & defense
	Topic 3. Malware analysis				

Today's session — Topic 1 (Windows OS security), its Lesson 1: Windows OS Services.

Lesson 1: Windows OS Services (Lecture)



Topic

Windows OS security



Session type

Lecture



Duration

2 hours

Learning questions

1 Service properties and functions

2 Starting and managing services



QUESTION 1

Service properties and functions

What is a Windows service?



Ordinary program

- ▶ Has a window on screen, you open it (Word, browser)
- ▶ You launch it when you are logged in



Service

- ▶ No window, runs invisibly in the background
- ▶ Starts automatically when the system boots

3 key differences between a service and an ordinary program:



No user needed

Runs even when no one is logged in



Automatic

Starts by itself as soon as Windows boots



Under a special account

With SYSTEM, LocalService or NetworkService rights

Why are services important for cybersecurity?



High privilege

Many services run at **SYSTEM** level — the highest right in Windows, even stronger than administrator. Taking over a service = controlling the whole system.



Persistence

Malware registers itself as a new **service** — so it restarts automatically even after the computer reboots (it gains persistence).



Privilege escalation

Through a misconfigured service (e.g. **unquoted service path**), an ordinary user can rise to SYSTEM rights.

QUESTION 1 · SERVICE PROPERTIES

Service Control Manager (SCM) — the services “controller”

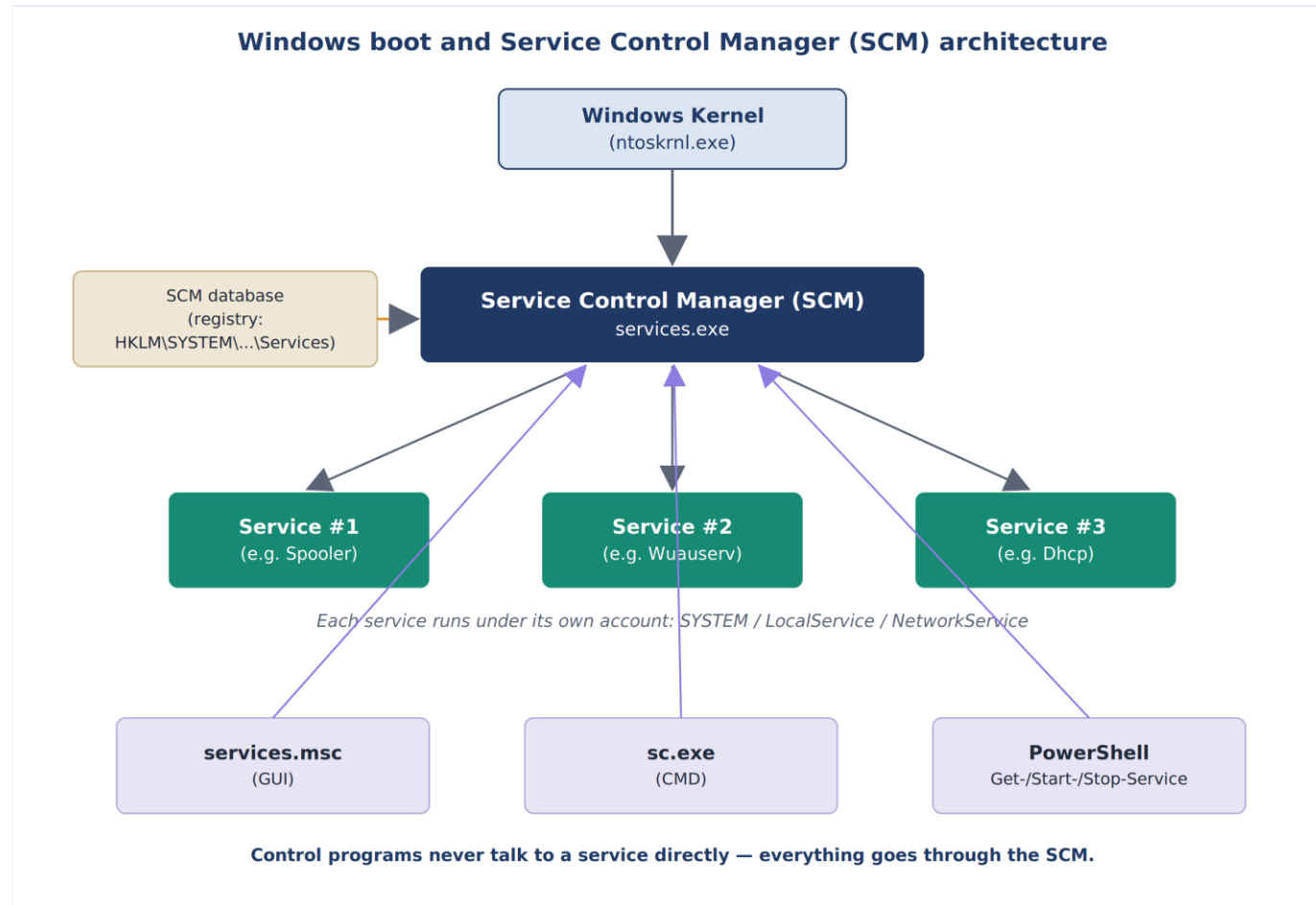


Fig. 1. Windows boot and SCM architecture

- ▶ The Kernel (**ntoskrnl.exe**) starts first.
- ▶ It launches **services.exe** — i.e. the SCM.
- ▶ SCM reads the full list of services **from the registry**.
- ▶ SCM starts them, monitors their state, and sets their permissions.
- ▶ **services.msc**, **sc.exe**, **PowerShell** do not talk to a service **directly** — everything goes through the SCM.

Service life cycle (Service States)

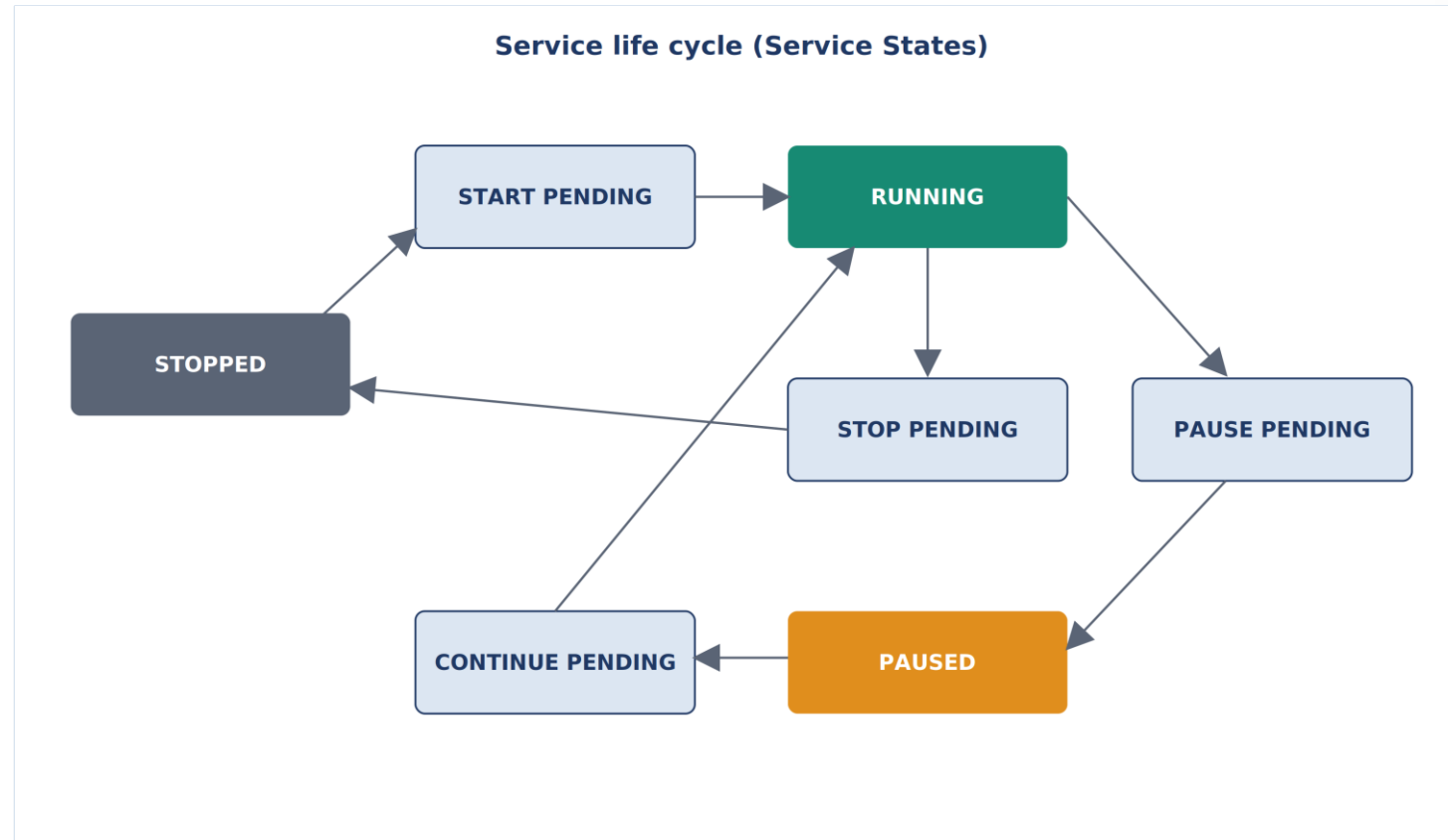


Fig. 2. Service states and transitions between them

Main states:

- ▶ **Stopped** — halted, not running
- ▶ **Start Pending** — in the process of starting
- ▶ **Running** — fully running
- ▶ **Paused** — temporarily suspended
- ▶ **Stop Pending** — in the process of stopping

Startup Types



Automatic

Starts immediately when the system boots

When:

Critical, always-on services (network, Update)



Automatic (Delayed)

Starts a bit later, after boot completes

When:

Needed, but not required immediately



Manual

Starts only when requested

When:

Rarely used ones (Print Spooler)



Disabled

Never starts at all

When:

Unused ones that may be risky

Under which account does a service run?

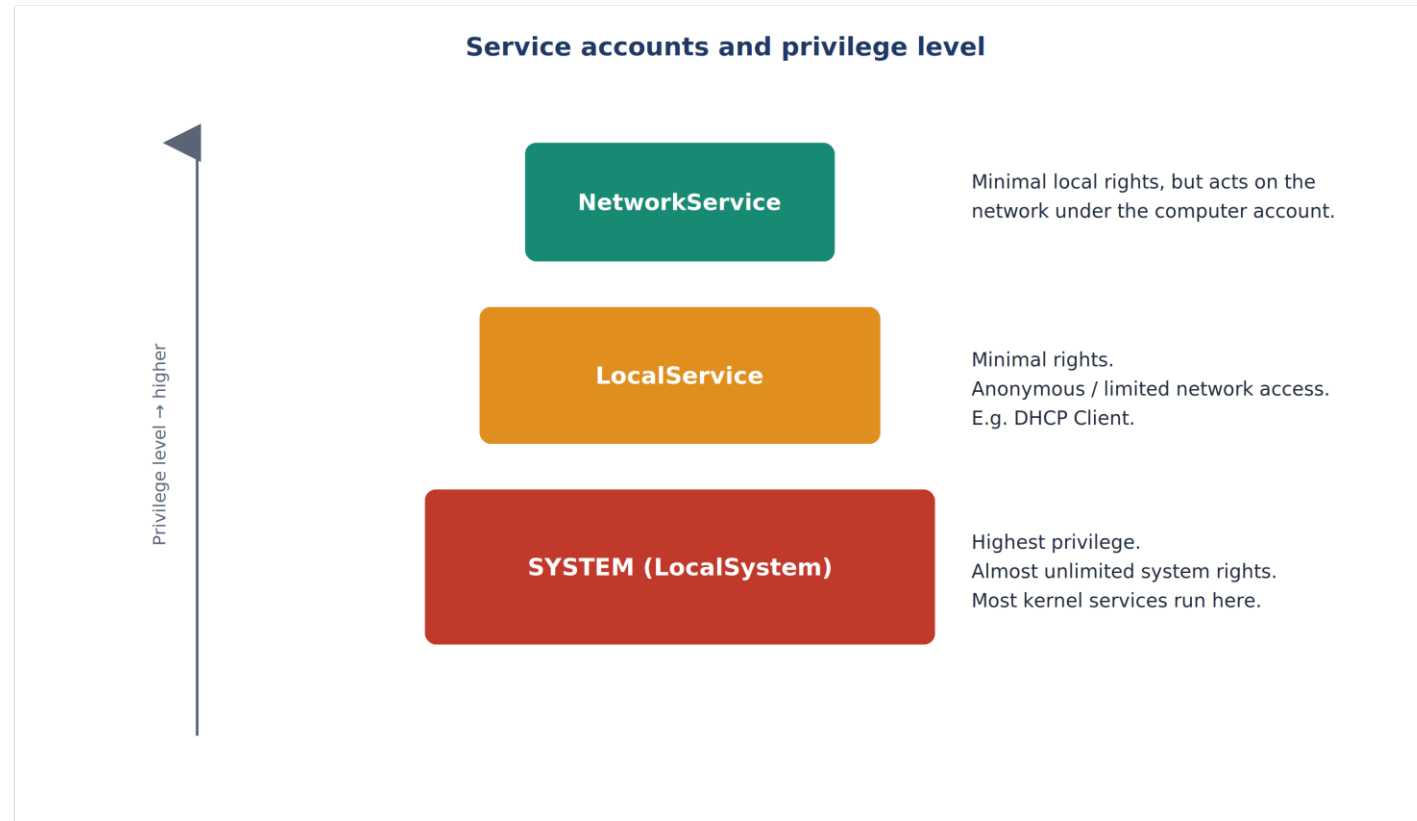


Fig. 3. Service accounts and their privilege level

SYSTEM — the highest, almost unlimited rights.

LocalService — minimal rights, limited network access.

NetworkService — minimal, but active on the network.

Security rule:

“**least privilege**” — the fewer rights a service is given, the safer the system.

A SYSTEM-level service is the most valuable prize for an attacker.



QUESTION 2

Starting and managing services

3 tools for managing services



GUI

services.msc

A graphical window used with the mouse. Shows all services, their state and the “Log On” column.



CMD

sc.exe

The classic command-line tool. Syntax: sc [command] [service_name] [parameters].



PowerShell

Get-Service

Modern, great for automation. Verb-Noun cmdlets: Get-Service, Start-Service.

QUESTION 2 · STARTING SERVICES

Via the GUI — services.msc

To open:

Win

+

R

→ services.msc → Enter

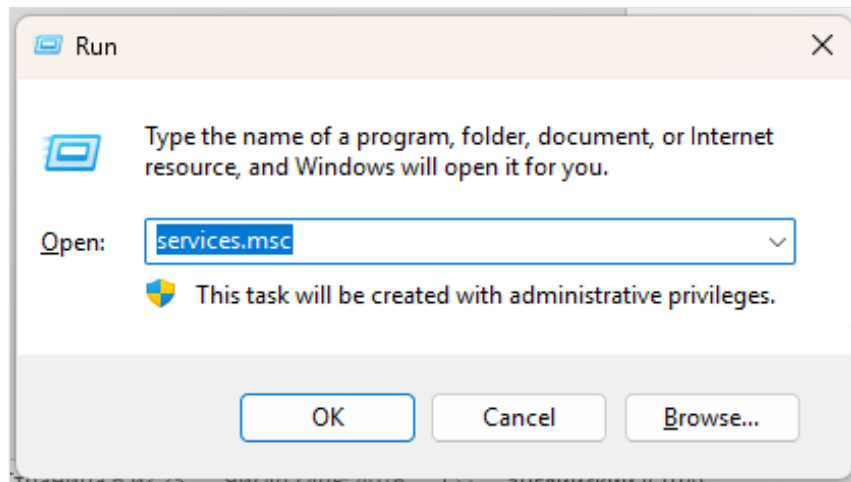


Fig. 4. services.msc in the Run box

- **Status** — Running / Stopped
- **Startup Type** — Automatic / Manual...
- **Log On As** — under which account

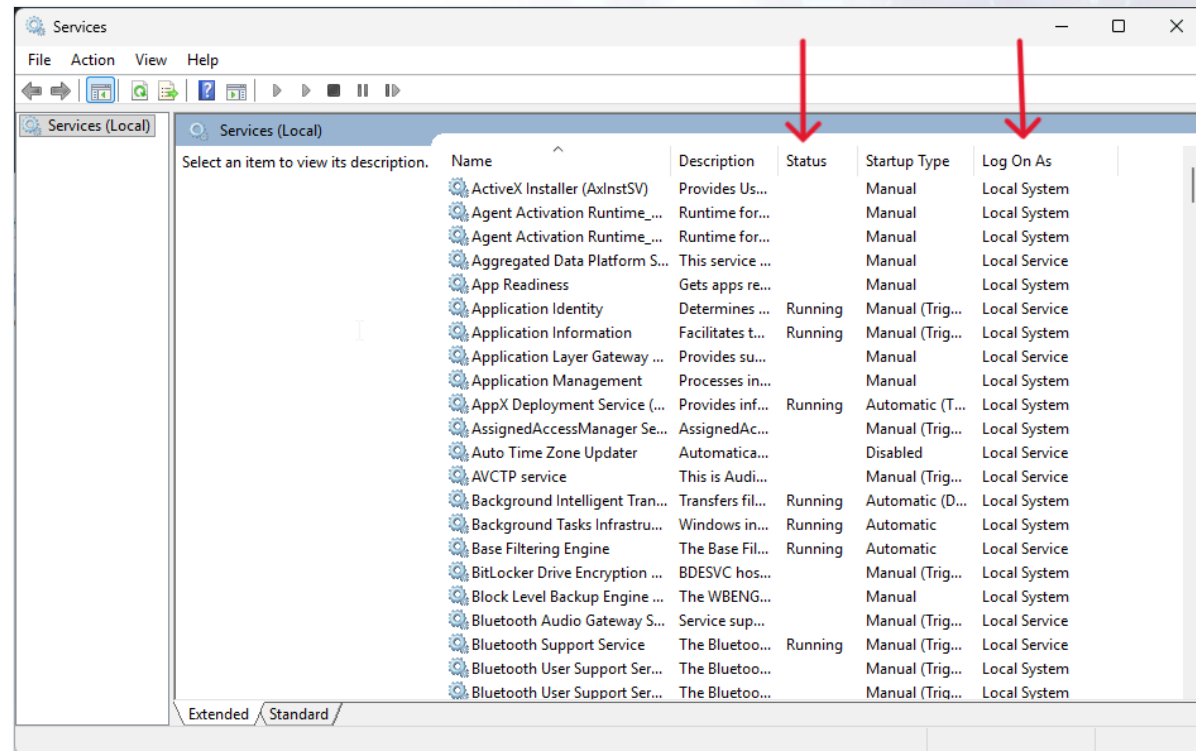


Fig. 5. Services window: Status, Startup Type and Log On As columns

Via CMD — sc.exe commands

Command	What it does	Example
sc query <name>	Shows the current state of a service	sc query Spooler
sc start <name>	Starts a service	sc start Spooler
sc stop <name>	Stops a service	sc stop Spooler
sc qc <name>	Full configuration (file path, account)	sc qc Spooler
sc config <name> start= disabled	Changes the startup type	sc config Spooler start= disabled



Note: leave a space after the “=” sign (**start= disabled**), but no space before it.

Via PowerShell — cmdlets

Cmdlet	Verb (what it does)	Noun (on what)
Get-Service	Get — retrieve, view	Service
Start-Service	Start — launch	Service
Stop-Service	Stop — halt	Service
Set-Service	Set — configure	Service



A cmdlet — a command that lives only inside PowerShell (not an ordinary **.exe** file).
It is always named in **Verb-Noun** form: **Get-Service**, **Start-Service**. That is why **Get-Service** does not work in CMD — only in PowerShell.

QUESTION 2 · PRACTICAL SIGNIFICANCE

sc qc — why it matters for security

```
> sc qc Spooler
```

```
BINARY_PATH_NAME :
```

```
C:\Windows\System32\spoolsv.exe
```

```
SERVICE_START_NAME : LocalSystem
```

- ▶ **BINARY_PATH_NAME** — the file path. If unquoted and contains spaces — it may be an **unquoted service path** vulnerability.
- ▶ **SERVICE_START_NAME** — shows which account the service runs under.
- ▶ In forensics and ethical hacking, **sc qc** is one of the first commands to check.



Blue Team: sets unnecessary services to **Disabled** to reduce the attack surface.



Red Team / Forensics: hunts for newly created or misconfigured services.

Terms (click for explanation):

? Unquoted service path

? MITRE ATT&CK

? cmdlet

Summary



Windows services — programs that run automatically in the background under special system accounts.



All services are managed by the Service Control Manager (SCM); the list is stored in the registry.



Three ways to manage them: GUI (services.msc), CMD (sc.exe), PowerShell (cmdlets).



A misconfigured service is a risk for privilege escalation and persistence.



This knowledge is the foundation for later topics (malware analysis, network monitoring).

Review questions — Lesson 1

1 How does a Windows service differ from an ordinary program?

2 What is the difference between the SYSTEM, LocalService and NetworkService accounts?

3 What functions does the SCM perform and where is it located?

4 Name the 4 startup types and explain when each is used.

5 Why can a misconfigured service lead to privilege escalation?

6 What is the difference between the sc query and sc config commands?

7 Why is checking services important in forensic analysis?



THANK YOU FOR YOUR ATTENTION!

Questions are welcome.

Next section — glossary of terms

Glossary — this section explains the following



Service

a program running in the background



Service Control Manager

the services dispatcher



Registry

the Windows settings database



Startup type

when it starts



Service accounts

SYSTEM/LocalService...



Unquoted service path

a dangerous vulnerability



MITRE ATT&CK

a knowledge base of attack techniques



cmdlet

a PowerShell command

Each term is explained in beginner-friendly language, with an example and a diagram.

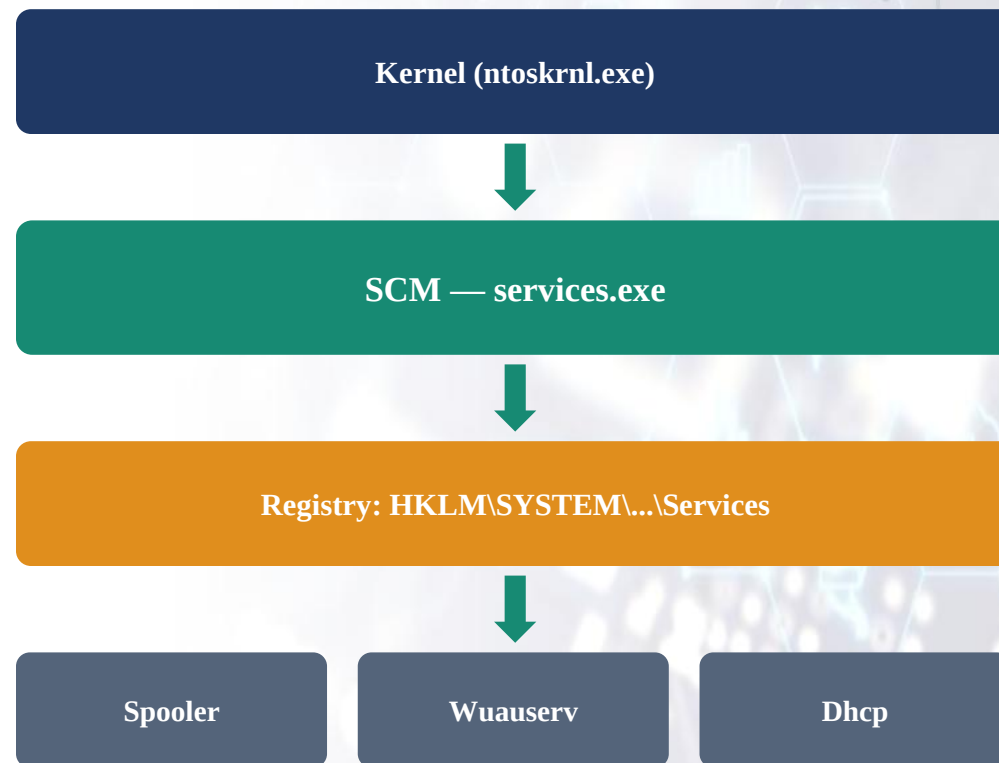
Service Control Manager (SCM)

In plain terms:

The SCM is the “boss” or dispatcher of all services. It runs as the services.exe file. When Windows boots, the kernel launches it; then the SCM reads the list in the registry and starts, stops and monitors each service.

Why it matters:

If you say “stop the Spooler”, the command does not go straight to the service — it goes to the SCM first, and the SCM carries it out. All control passes through the SCM.



Unquoted Service Path (vulnerability)

The problem: a service file path is unquoted and contains spaces. For example:

```
C:\Program Files\My App\service.exe
```

When it hits a space, Windows tries each of these in order:

C:\Program.exe

← if an attacker places a malicious file here, this runs first!

C:\Program Files\My.exe

← this is also tried

C:\Program Files\My App\service.exe

← the real, correct file (last of all)



Worst of all: because the service runs with **SYSTEM** rights, the malicious file also runs with SYSTEM rights — an ordinary user takes over the whole system. Fix: always put the path in quotes.

← Back to lesson

MITRE ATT&CK

In plain terms:

MITRE ATT&CK is a free, open knowledge base that organizes and catalogs the tactics and techniques attackers use in the real world (for example, “unquoted service path”).

Cybersecurity professionals use it to understand, detect and defend against attacks — it is the shared “vocabulary” of the field.



Example

Our vulnerability is coded in the base as:

T1574.009

Hijack Execution Flow: Path Interception by Unquoted Path

Source: attack.mitre.org

cmdlet (a PowerShell command)

In plain terms:

A cmdlet (pronounced “command-let”) is a small PowerShell command that does one specific job. It is not an ordinary .exe program — it runs only inside PowerShell.

A cmdlet is always named in “Verb-Noun” form: first what it does (verb), then on what (noun).

Note: **Get-Service** will not work if you type it in CMD — only in PowerShell.

Verb + Noun = cmdlet

Get-Service	Get (retrieve) · Service
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Start-Service	Start (launch) · Service
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Stop-Service	Stop (halt) · Service
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